

Chapter 4: Turning Effect of Force

Key Concepts

Like Parallel Forces — Forces acting in the same direction along parallel lines. Their resultant equals the sum of the forces and acts parallel to them in the same direction.

Unlike Parallel Forces — Forces acting in opposite directions along parallel lines. Their resultant equals the difference of the forces and acts in the direction of the larger force.

Resultant of Forces — A single force that produces the same effect as two or more forces acting together. For parallel forces, magnitude is the algebraic sum; for non-parallel forces, it follows the parallelogram law.

Torque or Moment of Force (τ) — The turning effect of a force about an axis. $\tau = F \times r \sin \theta$ where F is force and r is perpendicular distance (moment arm). SI unit: N m.

Clockwise and Anti-Clockwise Moments — A moment that rotates an object clockwise is a clockwise moment; anti-clockwise similarly. Convention: anti-clockwise is positive.

Principle of Moments — For a body in equilibrium, the sum of clockwise moments about any point equals the sum of anti-clockwise moments about the same point. $\sum \tau_{\text{clockwise}} = \sum \tau_{\text{anti-clockwise}}$.

Center of Mass — The point where the entire mass of a body is considered to be concentrated. For symmetrical bodies, it lies at the geometric center.

Center of Gravity — The point where the entire weight of a body acts. For uniform gravitational fields, it coincides with the center of mass.

Equilibrium — A body is in equilibrium when the net force and net torque acting on it are both zero.

First Condition of Equilibrium — The vector sum of all forces acting on a body is zero. $\sum \vec{F} = 0$. Ensures no translational acceleration.

Second Condition of Equilibrium — The sum of all torques acting on a body about any point is zero. $\sum \tau = 0$. Ensures no rotational acceleration.

Stable Equilibrium — Body returns to original position after a slight displacement. Center of gravity is lowest. Example: a ball in a bowl.

Unstable Equilibrium — Body moves further away from original position after a slight displacement. Center of gravity is highest. Example: a ball on a hilltop.

Neutral Equilibrium — Body stays in new position after displacement. Center of gravity remains at same height. Example: a ball on a flat surface.

Stability — The resistance of a body to toppling. Increased by: lowering center of gravity, widening the base, increasing mass.

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MCQ's

1. The turning effect of a force is called:

A. Work B. Torque C. Power D. Energy

Answer: B

2. SI unit of torque is:

A. N B. N/m C. N m D. J

Answer: C

3. Two forces of 5 N and 3 N act in the same direction on a body. Their resultant is:

A. 2 N B. 8 N C. 15 N D. 1.67 N

Answer: B

4. The principle of moments states that for equilibrium:

A. Sum of all forces is zero B. Clockwise moments = Anti-clockwise moments C. All forces must be equal D. Net force equals net torque

Answer: B

5. The center of gravity of a uniform rod lies at its:

Answer: B

6. A body is in stable equilibrium when its center of gravity is at the:

A. Highest position B. Lowest position C. Same height always D. Midpoint

Answer: B

7. Torque depends on force and:

A. Time B. Velocity C. Perpendicular distance D. Mass

Answer: C

8. First condition of equilibrium relates to:

A. Rotational motion B. Translational motion C. Both D. Neither

Answer: B

9. A boy balances a seesaw by shifting his position. This is an example of:

A. First condition of equilibrium B. Second condition of equilibrium C. Both D. None

Answer: B

10. Which of the following has the greatest stability?

A. A tall narrow vase B. A short wide vase C. A sphere D. A pencil on its tip

Answer: B

11. The moment arm of a force is the:

A. Length of the force vector B. Perpendicular distance from pivot to line of action C. Total length of the object
D. Distance from center of mass

Answer: B

12. A body is in neutral equilibrium when displaced, it:

A. Returns to original position B. Moves further away C. Remains in new position D. Oscillates

Answer: C

13. If two unlike parallel forces of 10 N and 4 N act on a body, the resultant is:

A. 14 N B. 6 N C. 40 N D. 2.5 N

Answer: B

14. A nut is easier to loosen with a longer spanner because:

A. Force increases B. Moment arm increases C. Mass increases D. Friction decreases

Answer: B

15. The second condition of equilibrium is also called the condition for:

A. Translational equilibrium B. Rotational equilibrium C. Dynamic equilibrium D. Static equilibrium

Answer: B

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Short Questions

Q1. Define torque or moment of force. Write its formula and SI unit.

Ans: Torque is the turning effect produced by a force acting on a body about an axis. It is given by $\tau = F \times d$ where F is the force and d is the perpendicular distance from the axis to the line of action of the force. Its SI unit is newton-meter (N m).

Q2. Differentiate between like and unlike parallel forces.

Ans:

- **Like parallel forces** act in the same direction along parallel lines. Their resultant equals the sum of the forces. Example: two people pushing a car forward.
- **Unlike parallel forces** act in opposite directions along parallel lines. Their resultant equals the difference of the forces. Example: two people pushing a car in opposite directions.

Q3. State the principle of moments.

Ans: According to the principle of moments, when a body is in equilibrium under the action of forces, the sum of clockwise moments about any point is equal to the sum of anti-clockwise moments about the same point.
Mathematically: $\sum \text{clockwise} = \sum \text{anti-clockwise}$.

Q4. What is center of gravity? How is it related to center of mass?

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This ensures the body does not accelerate linearly. If the net force is zero, the body is either at rest or moving with constant velocity.

Example: A book resting on a table. Weight downward is balanced by the normal reaction force upward.

Second Condition of Equilibrium (Rotational Equilibrium):

The sum of all torques acting on the body about any point is zero.

$$\sum \tau = 0$$

This ensures the body does not rotate.

Example: A balanced seesaw. The clockwise torque of one person equals the anti-clockwise torque of the other person.

A book on a table satisfies both conditions: it is not moving translationally ($\sum F = 0$) and it is not rotating ($\sum \tau = 0$).

Q3. Explain the principle of moments. How can it be verified experimentally?

Ans: The principle of moments states that for a body in equilibrium, the sum of clockwise moments about any point equals the sum of anti-clockwise moments about the same point.

Experimental Verification:

1. Take a meter rule and balance it at its center (50 cm mark) on a knife edge.
2. Hang a known weight W_1 on the left side at a distance d_1 from the pivot.
3. Hang another known weight W_2 on the right side and adjust its position until the rule balances horizontally.
4. Measure the distance d_2 from the pivot to W_2 .
5. Calculate the anti-clockwise moment = $W_1 \times d_1$ (if W_1 is on the left).
6. Calculate the clockwise moment = $W_2 \times d_2$.
7. It will be found that $W_1 \times d_1 = W_2 \times d_2$, verifying the principle.

The principle is widely applied in levers, balances, seesaws, and all systems involving rotational equilibrium.

Q4. What is center of mass and center of gravity? How does the position of center of gravity affect stability?

Ans:

Center of Mass: The point at which the entire mass of a body is considered to be concentrated. For a symmetrical body of uniform density, it lies at its geometric center.

Center of Gravity: The point at which the entire weight of a body acts. In a uniform gravitational field, it coincides with the center of mass.

Effect of Center of Gravity on Stability:

- A body with a **low center of gravity** is more stable because a larger torque is required to topple it. The body returns to its original position after displacement.
- A body with a **high center of gravity** is less stable because even a small displacement can cause it to topple.

Examples:

- Racing cars have a low center of gravity to prevent rolling over.
- Tall buildings have wide bases to keep the center of gravity within the base.
- A person carrying a heavy load on the head walks upright to keep the center of gravity aligned.
- A double-decker bus is less stable than a regular bus because its center of gravity is higher.

Q5. Describe the three types of equilibrium with examples from daily life.

Ans:

1. Stable Equilibrium:

- When displaced, the body returns to its original position.
- Center of gravity is at the lowest possible position.
- Example: A pendulum at rest, a ball in a concave bowl, a cone resting on its base.
- When displaced, the center of gravity rises, and gravity pulls it back down to the lowest point.

2. Unstable Equilibrium:

- When displaced, the body moves further away from its original position.
- Center of gravity is at the highest possible position.
- Example: A pencil balanced on its tip, a cone balanced on its vertex, a ball on a convex surface.

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Solution:

2 N weight at 20 cm mark.

Moment due to 2 N = $2 \times 0.3 = 0.6$ N m (clockwise)

Anti-clockwise moment = $3 \times d$

$$d = 0.6 / 3 = 0.2 \text{ m} = 20 \text{ cm}$$
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Solution:

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Solution:

8 N weight at one end: distance from pivot = 1 m

Let required weight = W at distance 1 m on the other side.

$$W = 8 \text{ N}$$